

Project Demonstration

Demonstration of Proposed Features

Date	03 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated, serving as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented/Partial/Pending)	Demonstrated (Yes/No)	Remarks
1	Soil & Climate Input Form	Collects N, P, K, temperature, humidity, pH, rainfall via validated input fields	Implemented	Yes	Core UI feature
2	Crop Prediction	Predicts the single best-suited crop using the trained Random Forest model	Implemented	Yes	99% test accuracy
3	Confidence Score Display	Shows the model's prediction confidence as a percentage	Implemented	Yes	

4	Top-3 Alternate Suggestions	Displays the next two most likely crops as backup options	Implemented	Yes	
5	Input Validation	Enforces sensible min/max ranges on every input field	Implemented	Yes	
6	Error Handling	Friendly error messages if the model file is missing or prediction fails	Implemented	Yes	

Feature Implementation Summary

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%